

		answer
4(a)(i)	$\lambda = \frac{c}{f} = \frac{3.0 \times 10^8}{10 \times 10^9} = 0.030 \text{ m}$	[1] for correct wavelength
4(a)(ii)	distance between transmitter and P = $\sqrt{500^2 + 300^2}$ = 583 m Intensity at P = $\frac{50}{4\pi(583)^2}$ = $1.17 \times 10^{-5} \text{ W m}^{-2}$	[1] for correct substitution [1] for correct answer
4(a)(iii)	$I \propto \frac{1}{r^2}$ $I \propto A^2$ $\therefore A \propto \frac{1}{r}$ $\frac{A_o}{A_p} = \frac{r_p}{r_o}$ $A_p = \frac{r_o}{r_p} A_o = \frac{500}{583} (15)$ = 12.8 cm $\approx 13 \text{ cm}$	[1] for substitution [1] for answer
4(b)	Let the amplitude of the microwaves after passing through A is A_o. The amplitude of microwaves after passing through B, $A_B = A_o \cos \theta$. Amplitude of microwaves after passing through C,	[1] for correct equation for Malus' law / amplitude

	$A_C = A_B \cos(90^\circ - \theta)$ $= A_o \cos \theta \sin \theta$ Since $I \propto A^2$, $I = kA_C^2$ $= k(A_o \sin \theta \cos \theta)^2$ $= (kA_o^2) \sin^2 \theta \cos^2 \theta$ $= I_o \sin^2 \theta \cos^2 \theta$ where k is the constant of proportionality.	[1] for $I \propto A^2$ [1] for correct working
5(a)	Zero electric field strengths in sphere A (between $x = 0$ and	[1]